

CATHOLIC JUNIOR COLLEGE JC2 PRELIMINARY EXAM 2008

MATHEMATICS

9740/01

Paper 1

WEDNESDAY

10 September 2008

3 hours

Additional materials: Answer Paper; List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your Name and Class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **ALL** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the paper, detach the cover page and fasten all your work securely with the cover page on top.

Name: _____

Class: _____

Question No	Marks	Question No	Marks
1		7	
2		8	
3		9	
4		10	
5		11	
6		12	
		Total	

1. A company specializes in making toy cars. It makes three types of toy cars, Toy Car A, Toy Car B and Toy Car C. The amount of materials required (in units) for each type of car is given below:

	Plastic	Rubber	Metal
Toy Car A	2	1	1
Toy Car B	3	4	5
Toy Car C	1	3	2
Total amt of material available	100	150	130

If all the materials available are used, how many cars of each type does the company make?

If the company makes a profit of \$1, \$2 and \$3 from the sale of each Toy Car A, B and C respectively, find the total profit it made if all the toy cars were sold. [4]

2. By using an algebraic method, solve the inequality $\frac{x-10}{x+2} > x-5$. [3]

Hence, solve the inequality $\frac{\sqrt{x}-11}{\sqrt{x}+1} > \sqrt{x}-6$. [3]

3. The product of the first three terms in a convergent geometric series is 1728. When the third term is decreased by 2, the three numbers will then form an arithmetic series.
- (i) Find the first three terms of the geometric series. [4]
 - (ii) State the condition for convergence and find the sum to infinity, S , of the geometric series. [2]
 - (iii) The sum of the first n terms of the arithmetic series is denoted by S_n . Find the values of n for which $5S_n$ exceeds S . [4]

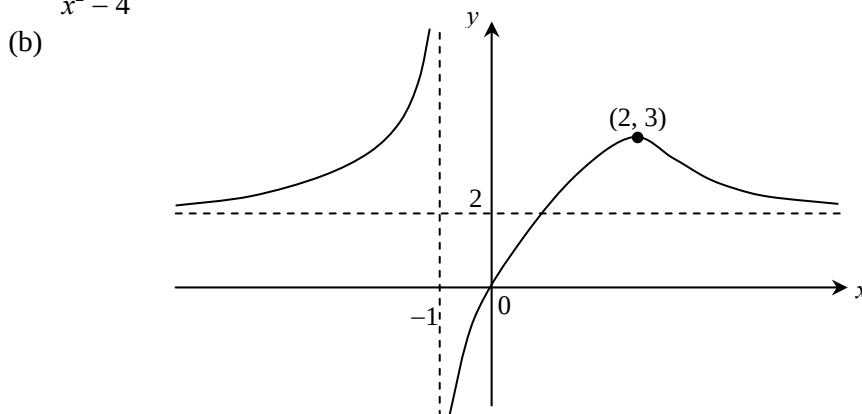
4. Functions f and g are defined by

$$f : x \rightarrow \frac{1}{1-e^{-x}}, \quad x \in \mathfrak{R}, x > 0$$

$$g : x \rightarrow -|1-x|, \quad x \in \mathfrak{R}, x > 0$$

- (i) With the aid of a diagram, show that f^{-1} exists. [2]
- (ii) Find f^{-1} in a similar form. [3]
- (iii) Only one of the composite functions fg and gf exists. Give a definition (including the domain) of the composite that exists and explain why the other composite does not exist. [3]
- (iv) The function g has an inverse if its domain is restricted to $x \geq k$. Find the smallest possible value of k and define the inverse function g^{-1} corresponding to this domain for g . [2]

5. The equation $x^3 - 6x + 1 = 0$ has a root, denoted by α , between $x = 0$ and $x = 1$. [1]
- (i) Find the value of α , correct to 5 decimal places. [1]
- (ii) A sequence of real numbers $x_1, x_2, x_3, \dots$ satisfies the recurrence relation $x_{n+1} = \frac{1}{6}(x_n^3 + 1)$ for $n \in \mathbb{Z}^+$. Use a calculator to determine the behaviour of the sequence for $x_1 = 0$. [2]
- (iii) Prove algebraically that, if the sequence in (ii) converges to α , then the sequence can be used to obtain the root α of the equation $x^3 - 6x + 1 = 0$. [2]
6. (a) Sketch the graph of $y = \frac{(x-4)^2}{x^2-4}$, showing clearly the asymptotes, the intersections with the axes and the coordinates of the turning points (if any). [4]
- Hence, or otherwise, state the range of values of m for which the equation $\frac{(x-4)^2}{x^2-4} - m = 0$ has no real roots. [1]



The diagram above shows the graph of $y = f(x)$.

On separate diagrams, sketch the graphs of

- (i) $y = f(3x - 1)$ [3]
- (ii) $y^2 = f(x)$ [3]
- (iii) $y = f'(x)$ [3]
- showing clearly in each case the axial intercepts, the asymptotes and the coordinates of the turning points.

7. A curve C has the following parametric equations:
 $x = k + \sin t$, $y = k \cos t$ where k is a fixed positive integer, $-\pi \leq t \leq \pi$.
- (i) Express $\frac{dy}{dx}$ in terms of k and t . [1]
- (ii) State the exact values of t at the points when the tangents are parallel to the y -axis and the points when the tangents are parallel to the x -axis. [2]
- (iii) The normal of the curve at the point where $t = \frac{\pi}{4}$ has a y -intercept of -1. Find the value of the constant k . [3]
- (iv) The normal intersects the curve at another point P. Using $k = 1$, find the intersection point. [3]

8. (a) Find $\int \sin^{-1} 2x \, dx$. [3]
- (b) Find $\int \frac{x}{x^2 - 2x + 3} \, dx$. [3]
- (c) Using the substitution $2x = \tan \theta$, find $\int \frac{2}{\sqrt{4x^2 + 1}} \, dx$. [3]
- (d) Evaluate $\int_1^4 |a^2 - x^2| \, dx$ where $1 < a < 4$, giving your answer in terms of a . [3]
9. (a) The complex number z is such that it has modulus 2 and argument $\frac{\pi}{2}$.
If $z = \frac{a}{1+i} + \frac{2b}{1-i}$, find the values of a and b , where $a, b \in \mathbb{R}$. [3]
- (b) The cubic equation $P(z) = 0$ has real coefficients.
If two of the roots are 1 and i , state the third root and find the equation in the form $Az^3 + Bz^2 + Cz + D = 0$. [3]
- (c) Solve the equation $iz^4 = -81$, expressing the solutions in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [4]
10. The variables x and y are related by the differential equation $\frac{dy}{dx} = 1 + 2x - \frac{y}{x^2}$.
Show that by means of the substitution $y = v + x^2$, the differential equation may be reduced to $\frac{dv}{dx} = -\frac{v}{x^2}$. [2]
Find the solution of the differential equation given that when $x = 1$, $y = 2$. [5]
11. Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, referred to an origin O .
Prove that the point Q whose position vector $\mathbf{q}$ is given by $\mathbf{q} = \lambda \mathbf{a} + (1 - \lambda)\mathbf{b}$ is collinear with A and B . Given that $\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix}$, find the two possible values of λ for which the angle between $\overrightarrow{OA}$ and $\overrightarrow{OQ}$ is 60° . [6]
12. (a) Sketch, on an Argand diagram, the locus of points representing the complex number z such that $|iz - 2| = |2 - z|$. Hence find the least value of $|z + 2|$. [3]
- (b) The conjugate of the non-zero complex number z is denoted by z^* . In an Argand diagram with origin O , the point P represents z and Q represents $\frac{1}{z^*}$.
(i) Find the ratio $OP : OQ$ in terms of $|z|$. [2]
(iii) Given that $w = z + \frac{1}{z^*}$, where $z = 2(\cos \theta + i \sin \theta)$, express the real and imaginary parts of w in terms of θ . [2]

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